

SEDAR 1: South Atlantic Red Porgy

Assessment Workshop

- The discrepancy between SC and NC aging is a major need that must be resolved, preferably before the next assessment. The SAW recommends that as soon as possible, the NC and SC investigators meet and share age readings techniques, to resolve the systematic discrepancies in age determinations, if possible. The SAW further recommends that research be undertaken that will accomplish verification of aging in red porgy.
- The protogeny of red porgy is a life history feature that complicates assessment and management. The SAW recommends that sampling for sex ratio at length be instituted in each fishery and that population sampling for sex ratio at length be continued by the MARMAP program. The SAW further recommends that research be instituted into assessment and population-projection methods that can make better use of sex-ratio data that exist now and that may exist in the future.
- Under many forms of management, considerable discarding of red porgy could be expected to occur. The SAW recommends that sampling programs be initiated to quantify discard rates, especially in the commercial fishery, where the discard mortality rate is believed higher, and to estimate discard mortality rates. The SAW recommends that research be instituted on management strategies that could reduce discard mortality and also research to illustrate the effects of discard mortality. The SAW also recommends that socioeconomic research be considered on educational measures to assist fishery participants in minimizing discard mortality and understanding the value of doing so.
- Fishery-independent data collected by the MARMAP program have served an important role in understanding the dynamics of this population, and the National Research Council has recommended that fishery-independent data play a more important role in stock assessment generally. However, the MARMAP sampling programs have been criticized by some as not having ideal extent, both in area coverage and in sampling intensity, for red porgy. The SAW recommends that the MARMAP program expand its coverage as needed.
- During the DW and SAW, it was noted that some incomplete, or misleading data have been entered in the NMFS general canvass data base. In particular, some data are available only under aggregated categories (e.g., porgies), even when accepted corrections to provide estimates of red porgy landings exist. The SAW recommends that state agencies contact and work with NMFS personnel maintaining the general canvass data base to make sure that data in that central data base are at the most disaggregated level possible and as accurate as possible.

The goal is that future red porgy assessment should be able to use data from the general canvass data base with confidence and without further corrections.

- A hook and line index of abundance should be developed for deeper water.
- The aging assumptions and the plus-group assumptions in the age structured model should be evaluated.
- Alternative assumptions about M should be evaluated.
- Sampling of catch by sex from commercial vessels should be initiated.
- Analyses to develop indices of abundance should consider the effects of unsuccessful effort.

Review Workshop

Research Recommendations

- Sampling for sex ratio is needed where protogeny is a concern; models and evaluations should incorporate this feature. Stock assessment scientists should discuss and develop methods to deal with these species. The implications of alternative assumptions about spawning stock definitions should be investigated.
- At-sea observers should be considered for monitoring discard and developing CPUE indices.
- Red porgy switch sex from females to males. The analytical tools and biological reference points do not take this into consideration. Implications of this are unknown and could have important affects on reference points and estimates of recovery.
- Concern was expressed that important information on the status of larger red porgy derived from deeper waters was not available as a separate index for inclusion in the assessment. It is recommended that further consideration be given to developing such indices from commercial and fishery independent data.
- Effective monitoring of stock recovery, especially under further fishing mortality reductions, will require information on discards.
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Process Recommendations

- The three step process (DW, SAW & SARC) proved to be very useful. It is recommended that more time be allocated between each of these steps. It would be helpful to have this incorporated into the Terms of Reference.
- If more than one stock is to be assessed per year, substantial additional resources must be provided. Additional funding will be necessary for NMFS and state participants.
- Participation of industry was a very important part at each step of the process. This practice should be continued.
- Priorities as to the stocks to be assessed need to be set.
- Having both NMFS and state scientists participating in the decision process for input data and assumptions for the model was very useful.
- Input from SARC participants other than on the panel was very useful. This will facilitate exchanges between the SAW and SARC participants.
- As well as peer review, the SARC was a useful forum for the exchange of technology and ideas.
- In future, the SARC will draft the Consensus Report at the meeting with a subsequent review.
- Improved technical support is required; printers, copiers, hard copy of drafts, LAN and other support.

CIE Consultant Recommendations

- Future SARCs should be larger; there was no buffer.
- More emphasis should be placed on systematic and structured comparison (figures and tables) with earlier assessments.
- The Chair was required to fill two roles; steering the meeting and as a technical reviewer.